XR-200 Centrifugal Pump - Equipment Overview

The XR-200 is a horizontal centrifugal pump used for transferring produced water and light crude in upstream field operations. Rated flow: 450 GPM at 150 ft head. Maximum operating pressure: 275 PSI. Maximum fluid temperature: 180 F. The unit consists of a cast steel volute casing, a single-stage enclosed impeller, a mechanical shaft seal, and an API 610 compliant baseplate. Motor: 75 HP, 3-phase, 460V, driven through a flexible coupling.

Section 4: Mechanical Seal Replacement Procedure

Tools required: 10mm and 13mm socket set, seal puller, torque wrench, clean rags, replacement mechanical seal kit (part no. XR200-MS-14).

Step 1: Isolate the pump from the system by closing the suction and discharge valves. Relieve all residual system pressure through the vent valve before proceeding.

Step 2: Lock out and tag out the motor power supply per site LOTO procedure.

Step 3: Remove the coupling guard and disconnect the coupling.

Step 4: Remove the seal housing bolts (8x M10) and slide the housing back along the shaft.

Step 5: Extract the old mechanical seal using the seal puller. Inspect the shaft sleeve for scoring or wear; replace the sleeve if wear exceeds 0.5mm.

Step 6: Clean the shaft and seal seat thoroughly with a lint-free cloth. Do not use abrasive materials on the seal seat surface.

Step 7: Install the new mechanical seal with the spring facing the impeller. Lubricate the shaft lightly with glycerin before sliding the seal into place.

Step 8: Reinstall the seal housing and torque the housing bolts to 35 Nm in a criss-cross pattern.

Step 9: Reconnect the coupling, reinstall the guard, remove lockout, and perform a slow-roll test before returning the pump to full operating pressure.

Section 7: Troubleshooting Guide

Symptom: Pump is leaking at the seal.
Likely cause: Worn or improperly seated mechanical seal, or a scored shaft sleeve.
Recommended action: Shut down the pump, follow the seal replacement procedure in Section 4. Inspect the shaft sleeve for wear during replacement.

Symptom: Pump is cavitating (loud rattling noise, unstable discharge pressure).
Likely cause: Insufficient suction pressure (NPSH), clogged suction strainer, or air ingress at the suction line.
Recommended action: Check suction strainer for blockage, verify suction valve is fully open, and inspect suction piping for air leaks. Do not run the pump dry.

Symptom: Motor trips on overload shortly after start-up.
Likely cause: Discharge valve closed on start-up, bearing failure, or impeller binding.
Recommended action: Confirm discharge valve is open before start-up. If the trip persists with the valve open, shut down and inspect bearings and impeller clearance.

All troubleshooting steps involving disassembly, electrical isolation, or pressurized components must be verified by qualified maintenance personnel before execution.

Section 9: Routine Maintenance Schedule

Daily: Visually inspect for leaks at the seal and flanges. Check discharge pressure gauge against baseline.

Weekly: Check motor and bearing temperatures with an infrared thermometer. Log vibration readings if a handheld vibration meter is available.

Monthly: Grease bearings per the lubrication chart (Section 10). Inspect coupling alignment and coupling element wear.

Every 6 months: Replace mechanical seal if vibration or minor weeping has been logged, even if not yet leaking. Inspect baseplate grouting for cracks.

Annually: Full pump teardown and inspection, including impeller wear ring clearance measurement and shaft runout check.

All maintenance recommendations in this manual are general guidance only and must be verified against site-specific procedures by qualified personnel before being carried out.